

Задача 4

8.4.12

$$\int \frac{dx}{x + \sqrt[3]{x^2}} = \left[\begin{array}{l} k = \text{НОК}(1; 3) = 3 \Rightarrow \\ \Rightarrow x = t^k = t^3 \Rightarrow \sqrt[3]{x^2} = \sqrt[3]{t^6} = t^2; \\ dx = d(t^3) = 3t^2 dt \end{array} \right] =$$

$$= \int \frac{3t^2 dt}{t^3 + t^2} = \int \frac{3t^2 dt}{t^2(t+1)} = 3 \int \frac{t^2 dt}{t^2(t+1)} = 3 \int \frac{dt}{t+1} =$$

$$= \int \frac{d(t+1)}{t+1} = 3 \int \frac{d(t+1)}{t+1} = 3 \ln|t+1| + C = \underline{3 \ln|\sqrt[3]{x} + 1| + C}.$$

8.4.13

$$\int \frac{\sqrt{x'}}{1 + \sqrt[4]{x^3}} dx = \left[\begin{array}{l} k = \text{НОК}(2; 4) = 4 \Rightarrow \\ \Rightarrow x = t^4 \Rightarrow \sqrt{x'} = t^2; \\ dx = d(t^4) = 4t^3 dt \end{array} \right] =$$

$$= \int \frac{t^2 \cdot 4t^3 dt}{1 + t^3} = \int \frac{4t^5 dt}{1 + t^3} = 4 \int \frac{t^5 dt}{1 + t^3} = \left[\begin{array}{l} u = 1 + t^3 \Rightarrow du = 3t^2 dt \\ \Rightarrow \frac{1}{3} du = t^2 dt; t^3 = u - 1 \end{array} \right] =$$

$$= 4 \int \frac{1}{3} \cdot \frac{(u-1)}{u} du = \frac{4}{3} \int du - \frac{4}{3} \int \frac{du}{u} = \frac{4}{3} u - \frac{4}{3} \ln|u| + C =$$

$$= \frac{4}{3} (1 + t^3) - \frac{4}{3} \ln|1 + t^3| + C = \underline{\frac{4}{3} (1 + \sqrt[4]{x^3}) - \frac{4}{3} \ln|1 + \sqrt[4]{x^3}| + C}$$

8.4.14

$$\int \frac{x + \sqrt[3]{x^2} + \sqrt[6]{x}}{x(1 - \sqrt[3]{x'})} dx = \left[\begin{array}{l} k = \text{НОК}(3; 6) = 6 \Rightarrow \\ \Rightarrow x = t^6 \Rightarrow \sqrt[3]{x^2} = t^4; \sqrt[6]{x} = t; \sqrt[3]{x'} = t^2; \\ dx = d(t^6) = 6t^5 dt \end{array} \right] =$$

$$= \int \frac{t^6 + t^4 + t}{t^6 - t^2} 6t^5 dt = 6 \int \frac{t^6(t^5 + t^3 + 1)}{t^2(t^4 - 1)} dt = 6 \int \frac{t^9 + t^7 + t^4}{t^4 - 1} dt =$$

$$= \left[\frac{t^9 + 0t^6 + t^7 + 0t^6 + 0t^5 + t^4 + 0t^3 + 0t^2 + 0t + 1}{t^5 - 0 - 0 - 0 - t^5} \cdot \frac{t^4 - 1}{t^5 + t^5 - t - 1} \right] =$$

$$\begin{array}{r} t^7 + 0 - t^5 \\ t^7 + 0 + 0 + 0 - t^5 \\ - t^5 + 0 - t^3 \\ - t^5 + 0 + t \\ \hline - t^4 \end{array}$$

$\frac{+1}{0}$

$$\begin{aligned}
 &= 6 \int \frac{(t^4 - t)(t^5 + t^3 - t - 1)}{t^6 - 1} dt = 6 \int (t^5 + t^3 - t - 1) dt = \\
 &= 6 \left(\int t^5 dt + \int t^3 dt - \int t dt - \int dt \right) = \\
 &= 6 \left(\frac{t^6}{6} + \frac{t^4}{4} - \frac{t^2}{2} - t \right) + C = \\
 &= t^6 + \frac{3t^4}{2} - 3t^2 - 6t + C = \\
 &= x + \frac{3\sqrt[3]{x^2}}{2} - 3\sqrt[3]{x} - 6\sqrt[6]{x} + C
 \end{aligned}$$

8.4.15

$$\int \frac{\sqrt{x} dx}{x - \sqrt[3]{x^2}} = \left[k = \text{HOK}(2; 3) = 6 \Rightarrow \right. \\
 \left. \begin{aligned} &\Rightarrow x = t^6 \Rightarrow \sqrt[3]{x^2} = t^4; \sqrt{x} = t^3; \\ &dx = d(t^6) = 6t^5 dt \end{aligned} \right] =$$

$$= \int \frac{t^3}{t^6 - t^4} 6t^5 dt = 6 \int \frac{t^8}{t^6 - t^4} dt = 6 \int \frac{t^2}{t^2 - 1} dt =$$

$$= \left[\begin{array}{c} t^4 + 0t^3 + 0t^2 + 0t + 0 \\ t^4 - 0 - t^2 \end{array} \middle| \frac{t^2 - 1}{t^2 + 1} \right. \\
 \left. \begin{array}{c} t^2 + 0 \\ t^2 - 1 \\ 1 \end{array} \right] =$$

$$6 \int \frac{(t^2 - 1)(t^2 + 1) + 1}{t^2 - 1} dt =$$

$$= 6 \int (t^2 + 1 + 1) dt = 6 \left(\int t^2 dt + \int dt + \int dt \right) =$$

$$= 6 \frac{t^3}{3} + 6t + 6t + C = 2t^3 + 12t + \underline{12\sqrt{x} + 12\sqrt[6]{x}} + C.$$

8.4.16

$$\int \frac{\sqrt{x} dx}{t + \sqrt{x}} = \left[k = 2 \Rightarrow \right. \\
 \left. \begin{aligned} &\Rightarrow x = t^2 \Rightarrow \sqrt{x} = t; \\ &dx = d(t^2) = 2t dt \end{aligned} \right] =$$

$$\begin{aligned}
 &= \int \frac{t^2 t^2 dt}{t+1} = 2 \int \frac{t^2}{t+1} dt = 2 \int \frac{t^2 + t + t}{t+1} dt = 2 \left(\int \frac{(t-1)(t+1)}{t+1} dt + \int \frac{dt}{t+1} \right) = \\
 &= 2 \left(\int t dt - \int dt + \int \frac{d(t+1)}{t+1} \right) = 2 \left(\frac{t^2}{2} - t + \ln|t+1| \right) + C = \\
 &= t^2 - 2t + 2\ln|t+1| + C = \\
 &= x - 2\sqrt{x} + 2\ln|\sqrt{x}+1| + C = \\
 &= \underline{x - 2\sqrt{x} + 2\ln(\sqrt{x}+1) + C}.
 \end{aligned}$$

8.4.17.

$$\int \frac{\sqrt{x} dx}{1-\sqrt{x}} = \left[\begin{aligned} k &= \text{HOK}(2,3) = 6 \Rightarrow \\ \Rightarrow x &= t^6 \Rightarrow \sqrt{x} = t^3; \sqrt{x} = t^3; \\ \Rightarrow dx &= 6t^5 dt \quad \underline{I} = \end{aligned} \right.$$

$$= \int \frac{t^3 6t^5 dt}{1-t^2} = 6 \int \frac{t^8}{1-t^2} dt = -6 \int \frac{t^8}{t^2-1} dt =$$

$$\begin{array}{r}
 \begin{array}{r}
 t^8 + 0t^7 + 0t^6 + 0t^5 + 0t^4 + 0t^3 + 0t^2 + 0t + 0 \\
 \hline
 t^8 - 0 - t^6 \\
 \hline
 t^6 + 0 + 0t^4 \\
 \hline
 t^6 - 0 - t^4 \\
 \hline
 t^4 + 0 + 0t^2 \\
 \hline
 t^4 - 0 - t^2 \\
 \hline
 t^2 + 0 + 0 \\
 \hline
 t^2 - 0 - 1 \\
 \hline
 1
 \end{array}
 \quad \left| \begin{array}{l} t^2-1 \\ t^6+t^4+t^2+1 \end{array} \right. \\
 \hline
 \underline{I} =
 \end{array}$$

$$= -6 \int \frac{(t^2-1)(t^6+t^4+t^2+1)+1}{t^2-1} dt = -6 \left(\int (t^6+t^4+t^2+1) dt + \int \frac{dt}{t^2-1} \right) =$$

$$= -6 \left(\int t^6 dt + \int t^4 dt + \int t^2 dt + \int dt + \int \frac{dt}{t^2-1} \right) =$$

$$= -6 \left(\frac{t^7}{7} + \frac{t^5}{5} + \frac{t^3}{3} + t + \frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| \right) + C =$$

$$= \underline{-\frac{6 \times \sqrt{x}}{7} - \frac{6 \sqrt{x^5}}{5} - 2\sqrt{x} - 6\sqrt{x} - 3 \ln \left| \frac{\sqrt{x}-1}{\sqrt{x}+1} \right| + C}.$$

8.4.18.

$$\int \frac{\sqrt{x+2}}{x} dx = \left[k=2 \rightarrow x+2=t^2 \rightarrow \right. \\ \left. \rightarrow x=t^2-2; dx=2t dt \right] =$$

$$= \int \frac{t \cdot 2t dt}{t^2-2} = 2 \int \frac{t^2 dt}{t^2-2} = 2 \int \frac{t^2-2+2}{t^2-2} dt = 2 \left(\int dt + 2 \int \frac{dt}{t^2-2} \right) =$$

$$= 2t + \frac{2}{\sqrt{2}} \cdot \ln \left| \frac{t-\sqrt{2}}{t+\sqrt{2}} \right| + C =$$

$$= 2\sqrt{x+2} + \frac{2}{\sqrt{2}} \cdot \ln \left| \frac{\sqrt{x+2}-\sqrt{2}}{\sqrt{x+2}+\sqrt{2}} \right| + C =$$

$$= 2\sqrt{x+2} + \sqrt{2} \cdot \ln \left| \frac{\sqrt{x+2}-\sqrt{2}}{\sqrt{x+2}+\sqrt{2}} \right| + C.$$

8.4.19

$$\int \frac{x dx}{\sqrt{x+1} + \sqrt[3]{x+1}} = \left[k = \text{HOK}(2;3) = 6 \Rightarrow \right. \\ \left. \Rightarrow x+1=t^6 \Rightarrow \sqrt{x+1}=t^3; \sqrt[3]{x+1}=t^2; x=t^6-1; \right. \\ \left. dx = d(t^6-1) = 6t^5 dt \right] =$$

$$= \int \frac{(t^6-1)6t^5 dt}{t^3+t^2} = 6 \int \frac{t^2(t^3-t^3)}{t^2(t+1)} dt = 6 \int \frac{t^5-t^3}{t+1} dt =$$

$$= \left[\frac{t^9+0t^8+0t^7+0t^6+0t^5+0t^4-t^3+0t^2+0t+0}{t^9-t^3} \right. \\ \frac{t^8+0t^7}{t^8-t^2} \\ \frac{t^7+0t^6}{t^7-t^0} \\ \frac{t^6+0t^5}{t^6-t^4} \\ \frac{t^5+0t^4}{t^5-t^2} \\ \frac{t^4-t^3}{t^4-t^3} \\ \left. \frac{0}{0} \right] =$$

$$= 6 \int \frac{(t-1)(t^8+t^7+t^6+t^5+t^4+t^3)}{t-1} dt =$$

$$= 6 \left(\int t^8 dt + \int t^7 dt + \int t^6 dt + \int t^5 dt + \int t^4 dt + \int t^3 dt \right) =$$

$$= 6 \left(\frac{t^9}{9} + \frac{t^8}{8} + \frac{t^7}{7} + \frac{t^6}{6} + \frac{t^5}{5} + \frac{t^4}{4} \right) + C =$$

$$= \frac{2(\sqrt[6]{x+1})^9}{3} + \frac{3(\sqrt[6]{x+1})^8}{4} + \frac{6(\sqrt[6]{x+1})^7}{7} + x + \frac{6(\sqrt[6]{x+1})^5}{5} + \frac{3(\sqrt[6]{x+1})^4}{2} + C =$$

$$= \frac{2}{3}(x+1)^{\frac{3}{2}} + \frac{3}{4}(x+1)^{\frac{4}{3}} + \frac{6}{7}(x+1)^{\frac{7}{6}} + x + \frac{6}{5}(x+1)^{\frac{5}{6}} + \frac{3}{2}(x+1)^{\frac{2}{3}} + C$$

8.4.20

$$\int \frac{dx}{(x+1)^{3/2} + (x+1)^{1/2}} = \left[k=2 \Rightarrow x+1=t^2 \Rightarrow \right. \\ \left. \Rightarrow (x+1)^{3/2}=t^3; (x+1)^{1/2}=t; \right. \\ \left. dx=d(t^2-1)=2tdt \right] =$$

$$= 2 \int \frac{tdt}{t^3+t} = 2 \int \frac{dt}{t^2+1} = 2 \arctan t + C =$$

$$= 2 \arctan \sqrt{x+1} + C$$

8.4.21

$$\int \frac{\sqrt{1+x}+1}{\sqrt{1+x}-1} dx = \left[k=2 \Rightarrow x+1=t^2 \Rightarrow \right. \\ \left. \Rightarrow \sqrt{x+1}=t; dx=d(t^2-1)=2tdt \right] =$$

$$= \int \frac{t^2+1}{t^2-1} 2tdt = 2 \left(\int \frac{t^2}{t^2-1} dt + \int \frac{tdt}{t^2-1} \right) = \left[u=t^2-1 \Rightarrow \right. \\ \left. \Rightarrow du=2tdt \right] =$$

$$= 2 \left(\frac{1}{2} \int \frac{u+1}{u} du + \frac{1}{2} \int \frac{du}{u} \right) = \int du + \int \frac{du}{u} + \int \frac{du}{u} =$$

$$= u + \ln|u| + \ln|u| + C = t^2-1 + 2\ln|t^2-1| + C =$$

$$= t^2 + 2 \ln |t^2 - 1| + C = x + 1 + 2 \ln |x + 1 - 1| + C = \\ = \underline{x + 2 \ln |x| + C}.$$

8.4.22

$$\int \frac{x-t}{\sqrt{2x-t}} dx = \left[k=2 \Rightarrow 2x-t = t^2; \right. \\ \left. \begin{aligned} \sqrt{2x-t} = t; x = \frac{t^2+t}{2}; \\ dx = d\left(\frac{t^2+t}{2}\right) = \frac{1}{2} \cdot 2t dt = t dt \end{aligned} \right] = \\ = \int \frac{t^2-t}{2t} t dt = \frac{1}{2} \int t^2 dt - \frac{1}{2} \int dt = \frac{1}{2} \cdot \frac{t^3}{3} - \frac{1}{2} \cdot t + C = \\ = \frac{t^3}{6} - \frac{t}{2} + C = \frac{t^3 - 3t}{6} + C = \frac{(2x-t)\sqrt{2x-t} - 3\sqrt{2x-t}}{6} + C = \\ = \frac{\sqrt{2x-t}(2x-t-3)}{6} + C = \underline{\underline{\frac{\sqrt{2x-t}(x-2)}{3} + C}}.$$

8.4.25

$$\int \frac{dx}{\sqrt{(x-1)^3(x-2)}} = \int \frac{\sqrt{x-1} dx}{\sqrt{(x-1)^4} \cdot \sqrt{(x-2)}} = \int \frac{dx}{(x-1)^2 \cdot \sqrt{\frac{x-2}{x-1}}} = \\ = \left[k=2 \Rightarrow \frac{x-2}{x-1} = t^2; \sqrt{\frac{x-2}{x-1}} = t \Rightarrow x-2 = t^2 x - t^2 \Rightarrow x - t^2 x = 2 - t^2 \Leftrightarrow \right. \\ \Rightarrow x(1-t^2) = 2-t^2 \Rightarrow x = \frac{2-t^2}{1-t^2} \Rightarrow x = \frac{t^2-2}{t^2-1}; \\ \left. dx = d\left(\frac{t^2-2}{t^2-1}\right) = \frac{2t}{(t^2-1)^2} dt \right] = \\ = \int \frac{1}{\left(\frac{t^2-2}{t^2-1} - 1\right)^2 \cdot t} \cdot \frac{2t}{(t^2-1)^2} dt = \left[\frac{t^2-2}{t^2-1} - 1 = \frac{t^2-2-t^2+1}{t^2-1} = \frac{-1}{t^2-1} \right] = \\ = \int \frac{(t^2-1)^2}{t} \cdot \frac{2t}{(t^2-1)^2} dt = 2 \int dt = 2t + C = \underline{\underline{2\sqrt{\frac{x-2}{x-1}} + C}}.$$

8.4.27
$$\int \frac{dx}{(1-x)\sqrt{1-x^2}} = \int \frac{dx}{(1-x)\sqrt{(1-x)(1+x)}} = \int \frac{dx}{(1-x)^2 \sqrt{\frac{1+x}{1-x}}} =$$

$= [k=2 \Rightarrow \frac{1+x}{1-x} = t^2; \sqrt{\frac{1+x}{1-x}} = t; 1+x = t^2 - t^2 x \Rightarrow x + t^2 x = t^2 - 1 \Rightarrow$

$\Rightarrow x(1+t^2) = t^2 - 1 \Rightarrow x = \frac{t^2 - 1}{t^2 + 1};$

$dx = d\left(\frac{t^2 - 1}{t^2 + 1}\right) = \frac{4t}{(t^2 + 1)^2} dt]$

$$= \int \frac{1}{\left(1 - \frac{t^2 - 1}{t^2 + 1}\right)^2 \cdot t} \cdot \frac{4t}{(t^2 + 1)^2} dt =$$

$$= \int \frac{(t^2 + 1)^2}{4t} \cdot \frac{4t}{(t^2 + 1)^2} dt = \int dt = t + C = \sqrt{\frac{1+x}{1-x}} + C$$

8.4.29

$$\int x^3 \cdot \sqrt{1+x^2} dx = \int x^3 \cdot (1+x^2)^{\frac{1}{2}} dx = [m=3, n=2, p=\frac{1}{2}]$$

1) $p = \frac{1}{2} \notin \mathbb{Z}$

2) $\frac{m+1}{n} = \frac{3+1}{2} = 2 \in \mathbb{Z}$

$\Rightarrow 1+x^2 = t^2 \Rightarrow x^2 = t^2 - 1$

$x = \sqrt{t^2 - 1}$

$\Rightarrow dx = d(\sqrt{t^2 - 1}) = \frac{t}{\sqrt{t^2 - 1}} dt]$

$$= \int (\sqrt{t^2 - 1})^3 \cdot t \cdot \frac{t}{\sqrt{t^2 - 1}} dt =$$

$$= \int (t^2 - 1) \cdot t^2 dt = \int (t^4 - t^2) dt = \int t^4 dt - \int t^2 dt =$$

$$= \frac{t^5}{5} - \frac{t^3}{3} + C = \frac{(\sqrt{1+x^2})^5}{5} - \frac{(\sqrt{1+x^2})^3}{3} + C$$

8.4.31

$$\int \frac{dx}{\sqrt{x}(1-\sqrt{x})^2} = \int x^{-\frac{1}{2}} (1-x^{\frac{1}{2}})^{-2} dx = [m = -\frac{1}{2}; n = \frac{1}{2}; p = -2,$$

$$1) p = -2 \in \mathbb{Z}$$

$$\Rightarrow k = \text{НОК}(2; 2) = 2 \Rightarrow x = t^2; x^{\frac{1}{2}} = t; x^{-\frac{1}{2}} = t^{-1};$$

$$\Rightarrow dx = 2t dt]$$

$$= \int \frac{2t dt}{t(1-t)^2} = 2 \int \frac{dt}{(1-t)^2} = \left[u = 1-t \Rightarrow du = -dt \right] = -2 \int \frac{du}{u^2} = \frac{2}{u} + C =$$

$$= \frac{2}{1-t} + C = \frac{2}{1-\sqrt{x}} + C$$

8.4.32.

$$\int x^5 \sqrt[3]{(1+x^3)^2} dx = \int x^5 (1+x^3)^{\frac{2}{3}} dx = [m=5; n=3; p=\frac{2}{3};$$

$$1) p = \frac{2}{3} \notin \mathbb{Z}$$

$$2) \frac{m+1}{n} = \frac{5+1}{3} = 2 \in \mathbb{Z};$$

$$\Rightarrow k = 3 \Rightarrow 1+x^3 = t^3 \Rightarrow x = \sqrt[3]{t^3-1};$$

$$\Rightarrow dx = \frac{t^2}{\sqrt[3]{(t^3-1)^2}} dt]$$

$$= \int (\sqrt[3]{t^3-1})^5 \cdot t^2 \cdot \frac{t^2}{\sqrt[3]{(t^3-1)^2}} dt = \int (t^3-1) \cdot t^4 dt =$$

$$= \int t^7 dt - \int t^4 dt = \frac{t^8}{8} - \frac{t^5}{5} + C =$$

$$= \frac{\sqrt[3]{(1+x^3)^8}}{8} - \frac{\sqrt[3]{(1+x^3)^5}}{5} + C$$

8.4.53

$$\int \frac{dx}{x^3 \cdot \sqrt[3]{2-x^3}} = \int x^{-3} (2-x^3)^{-\frac{1}{3}} dx = [m=-3; n=3; p=-\frac{1}{3};$$

$$1) p = -\frac{1}{3} \notin \mathbb{Z}$$

$$2) \frac{m+1}{n} = \frac{-3+1}{3} \notin \mathbb{Z}$$

$$3) \frac{m+1}{n} + p = \frac{-3+1}{3} - \frac{1}{3} = -1 \in \mathbb{Z} \rightarrow$$

$$\Rightarrow k=3 \Rightarrow 2x^{-3} - 1 = t^3; x^{-3} = \frac{t+1}{2}; x^3 = \frac{2}{t^3+1}; x = \sqrt[3]{\frac{2}{t^3+1}};$$

$$\Rightarrow dx = -\frac{\sqrt[3]{2} \cdot t^2}{(t^3+1)^{4/3}} dt$$

$$= \int \frac{t+1}{2} \cdot \left(2 - \frac{2}{t^3+1}\right)^{-\frac{1}{3}} \cdot \left(-\frac{\sqrt[3]{2} \cdot t^2}{(t^3+1)^{4/3}}\right) dt = -\frac{1}{2} \int t dt =$$

$$= -\frac{1}{2} \cdot \frac{t^2}{2} + C = -\frac{t^2}{4} + C = -\frac{\sqrt[3]{(2x^{-3}-1)^2}}{4} + C =$$

$$= -\frac{(x^3-2)^{2/3}}{4x^2} + C$$

8.4.54

$$\int \sqrt{x'} (1 + \sqrt{x'})^3 dx = \int \sqrt{x'} (1 + 3\sqrt{x'} + 3x + x\sqrt{x'}) dx =$$

$$= \int (x^{\frac{1}{2}} + 3x + 3x^{\frac{3}{2}} + x^2) dx = \int x^{\frac{1}{2}} dx + 3 \int x dx + 3 \int x^{\frac{3}{2}} dx + \int x^2 dx =$$

$$= \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + 3 \frac{x^2}{2} + 3 \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + \frac{x^3}{3} + C =$$

$$= \frac{2x\sqrt{x'}}{3} + \frac{3x^2}{2} + \frac{6x^2\sqrt{x'}}{5} + \frac{x^3}{3} + C$$

8.4.36

$$\int \frac{dx}{\sqrt{1-2x-x^2}} = \int \frac{dx}{\sqrt{2-(x+1)^2}} = [u = x+1; du = dx] =$$

$$= \int \frac{du}{\sqrt{2-u^2}} = \arcsin \frac{u}{\sqrt{2}} + C = \arcsin \frac{x+1}{\sqrt{2}} + C =$$

$$= \arcsin \frac{\sqrt{2}x + \sqrt{2}}{2} + C.$$

8.4.37

$$\int \frac{(x-2)dx}{\sqrt{x^2-10x+29}} = \int \frac{(x+5)+3}{\sqrt{x^2-10x+25+4}} dx =$$

$$= \int \frac{(x-5)dx}{\sqrt{x^2-10x+29}} + 3 \int \frac{dx}{\sqrt{(x-5)^2+4}} =$$

$$= \left[\begin{array}{l} 1) t = x^2 - 10x + 29 \Rightarrow dt = (2x - 10)dx; \frac{1}{2}dt = (x-5)dx \\ 2) (x-5) = y \Rightarrow dx = dy \end{array} \right] =$$

$$= \frac{1}{2} \int \frac{dt}{\sqrt{t}} + 3 \int \frac{dy}{\sqrt{y^2+4}} = \frac{1}{2} \cdot 2\sqrt{t} + 3 \ln|y + \sqrt{y^2+4}| + C =$$

$$= \sqrt{x^2-10x+29} + 3 \ln|(x-5) + \sqrt{(x-5)^2+4}| + C =$$

$$= \sqrt{x^2-10x+29} + 3 \ln|x-5 + \sqrt{x^2-10x+29}| + C.$$

8.4.38

$$\int \frac{3x-5}{\sqrt{x^2-4x+5}} dx = \int \frac{3(x-2)+1}{\sqrt{x^2-4x+4+1}} =$$

$$= 3 \int \frac{x-2}{\sqrt{x^2-4x+5}} dx + \int \frac{dx}{\sqrt{(x-2)^2+1}} = \left[\begin{array}{l} 1) t = x^2 - 4x + 5 \Rightarrow dt = (2x-4)dx \\ \Rightarrow \frac{1}{2}dt = (x-2)dx \\ 2) y = (x-2) \Rightarrow dy = dx \end{array} \right] =$$

$$= \frac{5}{2} \int \frac{dt}{\sqrt{t}} + \int \frac{dy}{\sqrt{y^2+1}} = \frac{5}{2} \cdot 2\sqrt{t} + \ln|y + \sqrt{y^2+1}| + C =$$

$$= 5\sqrt{x^2-4x+5} + \ln|(x-2) + \sqrt{(x-2)^2+1}| + C =$$

$$= 5\sqrt{x^2-4x+5} + \ln|x-2 + \sqrt{x^2-4x+1}| + C.$$

8.4.39

$$\int \frac{x+t}{\sqrt{2x-x^2}} dx = \int \frac{(x-t)+2}{\sqrt{2x-x^2}} dx = \int \frac{x-t}{\sqrt{2x-x^2}} dx + 2 \int \frac{dx}{\sqrt{2x-x^2}} =$$

$$= \int \frac{x-t}{\sqrt{2x-x^2}} dx + 2 \int \frac{dx}{\sqrt{-(x-t)^2+1}} = \left[\begin{array}{l} t = 2x-x^2 \Rightarrow dt = (2-2x)dx \\ \Rightarrow -\frac{1}{2}dt = (x-t)dx \\ t = x-t \Rightarrow dy = dx \end{array} \right] =$$

$$= -\frac{1}{2} \int \frac{dt}{\sqrt{t}} + 2 \int \frac{dy}{\sqrt{1-y^2}} = -\frac{1}{2} \cdot 2\sqrt{t} + 2 \cdot \arcsin y + C =$$

$$= -\sqrt{2x-x^2} + 2\arcsin(x-t) + C$$

8.4.40

$$\int \frac{\sqrt{1-x^2}}{x} dx = \int x^{-1} \cdot (1-x^2)^{\frac{1}{2}} dx = [m=-1; n=2; p=\frac{1}{2}]$$

$$1) p = \frac{1}{2} \notin \mathbb{Z}$$

$$2) \frac{m+1}{n} = \frac{-1+1}{2} = 0 \in \mathbb{Z};$$

$$\Rightarrow k=2 \Rightarrow 1-x^2 = t^2 \Rightarrow x = \sqrt{1-t^2}$$

$$\Rightarrow dx = -\frac{t dt}{\sqrt{1-t^2}} \quad \left[\right] =$$

$$= \int (\sqrt{1-t^2})^{-1} \cdot t \cdot \left(-\frac{t dt}{\sqrt{1-t^2}} \right) = - \int \frac{t^2 dt}{1-t^2} = \int \frac{t^2 dt}{t^2-1} = \int \frac{t^2-1+1}{t^2-1} dt =$$

$$= \int dt + \int \frac{dt}{t^2-1} = t + \frac{1}{2} \ln \left| \frac{t-1}{t+1} \right| + C =$$

$$= \sqrt{1-x^2} + \frac{1}{2} \ln \left| \frac{\sqrt{1-x^2} - 1}{\sqrt{1-x^2} + 1} \right| + C$$

8.4.41 $\int \sqrt{4-x^2} dx = [x=2\sin t \Rightarrow dx=2\cos t dt; t=\arcsin \frac{x}{2}] =$

$$= 2 \int \sqrt{4-(2\sin t)^2} \cdot \cos t dt = 2 \int \sqrt{4-4\sin^2 t} \cdot \cos t dt =$$

$$= 2 \int \sqrt{4(1-\sin^2 t)} \cdot \cos t dt = 2 \int \sqrt{4\cos^2 t} \cdot \cos t dt =$$

$$= 2 \int 2\cos t \cos t dt = 4 \int \cos^2 t dt = 4 \int \frac{1+\cos 2t}{2} dt =$$

$$= 2 \int dt + 2 \int \cos 2t dt = 2t + \frac{2}{2} \sin 2t + C =$$

$$= 2\arcsin \frac{x}{2} + \sin(2\arcsin \frac{x}{2}) = [\sin 2t = 2\sin t \cos t] + C =$$

$$= 2\arcsin \frac{x}{2} + 2 \sin(\arcsin \frac{x}{2}) \cos(\arcsin \frac{x}{2}) + C =$$

$$= 2\arcsin \frac{x}{2} + 2 \cdot \frac{x}{2} \cdot \sqrt{1-(\frac{x}{2})^2} + C =$$

$$= 2\arcsin \frac{x}{2} + x \cdot \frac{\sqrt{4-x^2}}{2} + C$$

8.4.42 $\int x \cdot \sqrt[5]{x-2} dx = \int x' \cdot (x'-2)^{\frac{1}{5}} dx = [m=1; n=1; p=\frac{1}{5}]$

1) $p = \frac{1}{5} \notin \mathbb{Z}$

2) $\frac{m+p}{n} = \frac{1+\frac{1}{5}}{1} = \frac{6}{5} \in \mathbb{Z}$

$$\Rightarrow k=5 \Rightarrow x-2=t^5 \Rightarrow x=t^5+2;$$

$$\Rightarrow dx = 5t^4 dt$$

$$= \int (t^5+2) \cdot (t^5)^{\frac{1}{5}} \cdot 5t^4 dt = 5 \int t^{10} dt + 10 \int t^5 dt =$$

$$= \frac{5 \cdot t''}{11} + \frac{10 \cdot t^6}{6} + C = \frac{5}{11} \cdot (\sqrt[5]{x-2})'' + \frac{10}{6} (\sqrt[5]{x-2})^6 + C =$$

$$= \frac{5}{11} \cdot (\sqrt[5]{x-2})'' + \frac{5}{3} \cdot (\sqrt[5]{x-2})^6 + C$$

8.4.43

$$\int \frac{dx}{x^2 \sqrt{x^2+1}} = \int x^{-2} (x^2+1)^{-\frac{1}{2}} dx = [m=-2; n=2; p=-\frac{1}{2};$$

$$1) p = -\frac{1}{2} \notin \mathbb{Z}$$

$$2) \frac{m+1}{n} = \frac{-2+1}{2} = -\frac{1}{2} \notin \mathbb{Z}$$

$$3) \frac{m+1}{n} + p = -\frac{1}{2} - \frac{1}{2} = -1 \in \mathbb{Z}$$

$$\Rightarrow k=2 \Rightarrow x^2+1=t^2 \Rightarrow x^2=t^2-1 \Rightarrow$$

$$\Rightarrow x^2 = \frac{1}{t^2-1} \Rightarrow x = \frac{1}{\sqrt{t^2-1}}$$

$$\Rightarrow dx = \frac{-t dt}{\sqrt{t^2-1} \cdot (t^2-1)} =$$

$$= \int (t^2-1) \cdot \sqrt{\frac{1}{t^2-1} + 1} \cdot \left(-\frac{t dt}{\sqrt{t^2-1} \cdot (t^2-1)} \right) = -\int dt =$$

$$= -t + C = -\sqrt{x^2+1} + C = -\sqrt{\frac{1+x^2}{x^2}} + C$$

$$= -\frac{\sqrt{1+x^2}}{x} + C$$